

Fixed Points of a Random Permutation

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Publication page. The canonical page for this writing, including all three editions and their download links, is:

<https://opuscula.ritsuka.moe/euler-derangement-cloze/>

PDF editions.

[paper-en.pdf](#) [paper-en-revised.pdf](#) [paper-zh.pdf](#)

Source repository. The L^AT_EX sources, compiled PDFs, and edition notes are available at <https://github.com/Ritsuka314/opuscula>.

Purpose. The reconstruction was prepared solely for entertainment and educational purposes. It is unofficial, is not presented as an author-approved transcript, and makes no claim of affiliation with the video creator or platform.

The three editions.

1. **Reproduced English edition.** A close typesetting of the four-page English note visible in the video. It intentionally retains the source's nonstandard wording, omissions, notation, and induction argument.
2. **Revised English edition.** A corrected, self-contained mathematical report. It defines the model precisely, proves the derangement formula by inclusion–exclusion, gives a unified probability distribution, and proves the expectation with indicator variables.
3. **Revised Chinese edition.** A polished Chinese counterpart to the revised English report. It follows the same definitions, results, and proofs rather than translating the reproduced wording literally.

Edition in this file: Revised English edition
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This edition is a corrected, self-contained report based on the mathematical question in the source video.
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Fixed Points of a Random Permutation

Expected Value and Distribution

Abstract

Let a permutation of n distinct objects be chosen uniformly at random, and let X denote its number of fixed points. This report first proves directly that $\mathbb{E}[X] = 1$ for every positive integer n . It then develops the derangement formula step by step using inclusion–exclusion and derives the exact distribution of X . The result models, for example, the number of correct answers obtained when n distinct choices are assigned uniformly at random to n distinct blanks.

1 Problem setup

For a positive integer n , write

$$[n] = \{1, 2, \dots, n\}.$$

Let $\sigma : [n] \rightarrow [n]$ be a permutation chosen uniformly from the $n!$ elements of the symmetric group S_n . An index $r \in [n]$ is a *fixed point* of σ if $\sigma(r) = r$. Define

$$X = |\{r \in [n] : \sigma(r) = r\}|.$$

Our main question is the expected value of X ; afterward we determine its entire probability mass function as a stronger result.

2 Expected number of fixed points

Theorem 1. *For every $n \geq 1$,*

$$\mathbb{E}[X] = 1.$$

Proof. For each $r \in [n]$, define the indicator variable

$$I_r = \begin{cases} 1, & \sigma(r) = r, \\ 0, & \sigma(r) \neq r. \end{cases}$$

Then $X = \sum_{r=1}^n I_r$. Of the $n!$ permutations, exactly $(n-1)!$ fix a specified index r , so

$$\Pr(\sigma(r) = r) = \frac{(n-1)!}{n!} = \frac{1}{n} \quad \text{and} \quad \mathbb{E}[I_r] = \frac{1}{n}.$$

By linearity of expectation,

$$\mathbb{E}[X] = \sum_{r=1}^n \mathbb{E}[I_r] = n \cdot \frac{1}{n} = 1.$$

The indicator variables need not be independent for this argument. □

Equivalent double-counting view. Count the pairs $(\sigma, r) \in S_n \times [n]$ for which $\sigma(r) = r$. For each of the n choices of r , exactly $(n - 1)!$ permutations fix r , so there are $n(n - 1)! = n!$ such pairs. Dividing this total number of fixed-point incidences by the $n!$ permutations again shows that the average number of fixed points is 1.

3 Derangements and the exact distribution

The expectation proof does not require the distribution of X . To obtain the stronger result $\Pr(X = k)$ for every k , we first count permutations having no fixed points.

3.1 Counting derangements

A *derangement* is a permutation with no fixed points. Let D_m denote the number of derangements of m objects, with the convention $D_0 = 1$.

Theorem 2 (Derangement formula). *For every integer $m \geq 0$,*

$$D_m = m! \sum_{j=0}^m \frac{(-1)^j}{j!}.$$

Consequently, a uniformly random permutation of m objects is a derangement with probability

$$\frac{D_m}{m!} = \sum_{j=0}^m \frac{(-1)^j}{j!}.$$

Proof. For $m = 0$, the single empty permutation is a derangement, and the formula gives $D_0 = 1$. Now suppose $m \geq 1$, and take the full set S_m of all $m!$ permutations as the universe. For each $r \in [m]$, let

$$A_r = \{\sigma \in S_m : \sigma(r) = r\};$$

thus A_r is the set of permutations that fix r . A permutation is a derangement precisely when it lies in none of $A_1, \dots, A_m$, so

$$D_m = \left| \bigcap_{r=1}^m \overline{A_r} \right|.$$

Inclusion–exclusion begins with all $m!$ permutations, subtracts those that fix each specified element, adds back those subtracted twice because they fix two specified elements, and continues with alternating signs:

$$\begin{aligned} D_m &= |S_m| - \sum_{r=1}^m |A_r| + \sum_{1 \leq r < s \leq m} |A_r \cap A_s| - \dots \\ &\quad + (-1)^m |A_1 \cap \dots \cap A_m|. \end{aligned}$$

To evaluate these terms, choose a subset $J \subseteq [m]$ with $|J| = j$. If every element of J must be fixed, the remaining $m - j$ elements may be permuted arbitrarily. Hence

$$\left| \bigcap_{r \in J} A_r \right| = (m - j)!.$$

When $J = \emptyset$, the intersection is understood to be the whole universe S_m , so this statement also includes the $j = 0$ term. There are $\binom{m}{j}$ choices for J . Grouping the inclusion–exclusion terms according to j therefore gives

$$\begin{aligned} D_m &= \sum_{j=0}^m (-1)^j \binom{m}{j} (m-j)! \\ &= m! \sum_{j=0}^m \frac{(-1)^j}{j!}. \end{aligned}$$

For completeness, the alternating sum keeps exactly the desired permutations. A permutation with $q \geq 1$ fixed points occurs in $\binom{q}{j}$ of the terms involving j prescribed fixed points, so its net coefficient is

$$\sum_{j=0}^q (-1)^j \binom{q}{j} = (1-1)^q = 0.$$

A derangement has $q = 0$ and remains counted once. This also explains directly why inclusion–exclusion works here.

Dividing by the total number $m!$ of permutations proves the probability formula. $\square$

For example,

$$D_3 = 3! - \binom{3}{1}2! + \binom{3}{2}1! - \binom{3}{3}0! = 6 - 6 + 3 - 1 = 2.$$

In one-line notation, the two derangements are $[2, 3, 1]$ and $[3, 1, 2]$.

When $m \geq 2$, the terms for $j = 0$ and $j = 1$ cancel. Thus the same probability may also be written as

$$\sum_{j=2}^m \frac{(-1)^j}{j!},$$

which is the form used in the reproduced note.

3.2 Distribution of the number of fixed points

Theorem 3. For $k = 0, 1, \dots, n$,

$$\Pr(X = k) = \frac{\binom{n}{k} D_{n-k}}{n!} = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

Proof. First choose the k indices that will be fixed; this can be done in $\binom{n}{k}$ ways. The remaining $n - k$ indices must then be deranged, which can be done in D_{n-k} ways. Hence exactly $\binom{n}{k} D_{n-k}$ permutations have k fixed points. Dividing by $n!$ and substituting the derangement formula yields

$$\begin{aligned} \Pr(X = k) &= \frac{\binom{n}{k} D_{n-k}}{n!} \\ &= \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}. \end{aligned}$$

$\square$

The unified formula includes the two boundary cases:

$$\Pr(X = n) = \frac{1}{n!}, \quad \Pr(X = n - 1) = 0.$$

The second identity holds because a permutation cannot move exactly one element while fixing every other element.

4 Conclusion

The direct indicator argument shows that a uniformly random permutation has one fixed point on average. The stronger counting argument gives the exact distribution

$$\Pr(X = k) = \frac{\binom{n}{k} D_{n-k}}{n!},$$

where the derangement number D_{n-k} is obtained by inclusion–exclusion. Therefore, when n distinct answers are placed uniformly at random into n distinct blanks, one answer is correct on average. This is an expectation statement: an individual attempt may have no correct answers, one correct answer, or several.